Final Combined Financial Dataset: Tusheti Connectivity Project

1. High-Level Budget Summary (Global Overview)

This summary outlines the global financial allocations for the project. Per the **draft budget 3.0**, the total project envelope is **$121,022**. This figure represents the comprehensive implementation scope, encompassing **Phase 1: Tusheti Connectivity** (itemized below) and the planned **Phase 2: Khevsureti Expansion** (including Arkhoti and Barisakho regions).

Category	Allocation (USD)	Description/Notes
Solar Energy	$26,601	Panels, batteries (Phase 1 & 2), and power management systems.
Masts	$9,762	Physical infrastructure, main masts, and gap-filler structures.
Radio Equipment	$60,000	Core network hardware (AirFiber, PowerBeam, LiteBeam, etc.).
Tools	$7,101	Installation hardware, field technical kits, and maintenance tools.
Operational	$6,786	Field logistics, transportation, food, and accommodation.
Fees/Honorariums	$10,772	Engineering personnel, project design, and local operators.

Total Project Budget	**$121,022**	**Combined total for full regional implementation (Phase 1 & 2).**

2. Itemized Category Breakdown: Radio & Network Equipment

The following inventory details the technical core of the Tusheti backbone and local distribution nodes. In response to field relief analysis, sectoral antennas were upgraded to the LBE-5AC-16-120 model for improved coverage.

Item Name/Model	Unit Price ($)	Quantity	Total Cost ($)
AF-5x, 5GHz (AirFiber)	$470.00	4	$1,880.00
RD-34db Dish (34dBi)	$357.00	4	$1,428.00
PowerBeam PBE-5AC-620	$272.00	4	$1,088.00
LBE-5AC-16-120 (Sectoral Upgrade)	$115.00	14	$1,610.00
RB750P-PBr2 (PowerBox)	$90.00	3	$270.00
Sector Antenna AM-5G19-120	$155.00	7	$1,085.00
LiteBeam M5, 23 dBi	$59.00	12	$708.00
Rocket M5	$95.00	4	$380.00

RAD-3RD RocketDish Radome (3ft)	$135.00	3	$405.00
AF-5G-OMT-S45 Conversion Kit	$42.00	4	$168.00
6-Port Gigabit POE Injector	$85.00	6	$510.00
LAN POE Cable (Cat5, 100% Cu)	$142.00	1	$142.00

3. Itemized Category Breakdown: Solar Energy & Power Infrastructure

To maintain 12-month operational stability, especially during the high-snow winter months, the network topology was modified to provide **100W/12V** combined power per mast (utilizing 4x100W panels to feed the POE-BOX8-G system at 24V).

Component	Unit Cost	Quantity	Subtotal
Solar Panels (100W)	$285.00	20	$5,700.00
High-Capacity Batteries (Actual Spend)	$155.70	10	$1,557.00
Charge Controller (Nippotec)	$25.00	7	$175.00
Steel Construction for Solar Panels	$470.00	5	$2,350.00
Outdoor Rack Mount	$155.00	5	$775.00
High Power Cable (2.5mm/1.8mm)	$2.70	100	$270.00

4. Itemized Category Breakdown: Masts, Structures, and Fasteners

Structural infrastructure is engineered for high-altitude resilience. Technical drawings (tusheti_tower-01/02) specify iron pipe sections with anchor plates and concrete foundations.

- **Height:** 3.5 Meters
- **External Diameter:** 76mm
- **Wall Thickness:** 3mm
- **Foundation:** M300 Concrete (1/8 Cubic Meter per base)

Item	Unit Price ($)	Quantity	Total Cost ($)
Antenna Tower Anchors (M10 Thread)	$17.00	64	$1,088.00
Anchors and Stretch Marks (Tros)	$380.00	1	$380.00
Main Tower Sections (5-Site Deployment)	$110.00	5	$550.00
Bolts, Nuts, and Washers	$27.00	1	$27.00

5. Operating Expenses (OPEX) and Logistics

Logistical calculations cover the transportation of approximately 400-500kg of cargo per trip and regional engineering support.

Expense Type	Rate/Basis	Units	Total
Fuel (Oil/Diesel)	$1.20	540	$648.00

Food (Engineering Team)	$112.00	8	$896.00
Cargo Transportation (Truck)	$275.00	2	$550.00
Car Rental (Delica L300 4x4)	$115.00	9	$1,035.00
Guesthouse Accommodation (Diklo)	Flat Rate	3	$735.00
Guesthouse Accommodation (Omalo)	Flat Rate	4	$980.00
Unexpected Expenses (Contingency)	Lump Sum	1	$1,900.00

6. Salaries, Honorariums, and Professional Fees

A 20% Income Tax is applied to all gross personnel figures. The following includes all technical and local staff required for Phase 1.

Role/Individual	Daily Rate ($)	Duration (Days)	Gross Total ($)
Project Director (U.Seturi)	$78.00	19	$1,482.00
Project author (K. Stalinsky)	$147.00	10	$1,470.00
IT Engineer (Giorgi Kirvalidze)	$115.00	13	$1,495.00
Tower Designer (Murman Charelidze)	$94.00	8	$752.00
Technical Engineer (A. Giorganashvili)	$97.00	10	$970.00

Support Engineer (G. Matchavariani)	$9.00	97	$873.00
Local Operator (G. Kitidze)	$58.00	8	$464.00
Local Operators (2 persons)	$37.00	21	$777.00
Driver (Field Logistics)	$47.00	7	$329.00
Network Operator (G. Konashvili)	$72.00	2	$144.00
Income Tax (20%)	**-**	**-**	**$1,751.20**

7. Technical Tools and Field Equipment

Checklist for field readiness and tower installation.

- [ ] **Power Inverter (1200-1500W, 24V-220V):** $463.00
- [ ] **High Power Driver-Drill:** $517.00
- [ ] **Screwdriver/Drill Driver Kit (13/17/24mm):** $370.00
- [ ] **Hammers and SDS Shank Drill Bit Set:** $290.00

8. Site-Specific Equipment Allotment (Network Topology Mapping)

Rusipiri (Output)

- 1x AF-5x (5GHz)
- 1x RD-34db Dish
- 1x RAD-3RD RocketDish Radome
- 1x AF-5G-OMT-S45 Conversion Kit
- 1x RB2011UiAS-IN

Abano (Bridge)

- 2x AF-5x (5GHz)
- 2x RD-34db Dish
- 1x RAD-3RD RocketDish Radome
- 2x AF-5G-OMT-S45 Conversion Kit
- 1x PowerBeam PBE-5AC-620
- 1x 6-Port Gigabit POE Injector
- 1x Battery Charge Controller

Chiglaurta (Basic Point)

- 3x PowerBeam PBE-5AC-620 (Links: Abano/Diklo/Koklata)
- 3x LBE-5AC-16-120
- 2x 6-Port Gigabit POE Injector
- 1x RB750P-PBr2 (PowerBox)

Diklo (Bridge)

- 2x PowerBeam PBE-5AC-620
- 2x LBE-5AC-16-120
- 1x 6-Port Gigabit POE Injector
- 1x RB750P-PBr2 (PowerBox)

Makratela (Repeater)

- 2x PowerBeam PBE-5AC-620
- 3x LBE-5AC-16-120
- 1x 6-Port Gigabit POE Injector
- 1x RB750P-PBr2 (PowerBox)

Koklata

- 2x PowerBeam PBE-5AC-620
- 4x LBE-5AC-16-120
- 1x 6-Port Gigabit POE Injector
- 1x RB750P-PBr2 (PowerBox)

9. Key Financial Observations

- **Procurement Savings:** Secured a recurring project discount of 5.4% - 5.6% through LTD Freenet via Intellcom Group.
- **Power Optimization:** While 2,400 was originally planned for solar batteries, actual expenditure was optimized to **1,557**.

- **Infrastructure Reallocation:** A surplus of $1,695 from the panel mount budget was identified. Per PM directive, 10% of this was used for custom-made clips and bolts to replace unavailable factory components.
- **Thunder Protection Integration:** Following the hardware failures experienced by Mobitel in the region, the project procured additional **Andeli Thunder Protection Systems** to safeguard against high sulfide/ferrite metal lightning attraction on the ridges.
- **Capacity Increase:** Replacing sector antennas (155/unit) with LBE units (115/unit) allowed for a doubling of sectors (7 to 14), significantly increasing subscriber potential within the same budget envelope.

10. Administrative Reference Data

Organization Name:
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